

Resolving nanosecond kinetics of the optical membrane potential in pulsed electric fields

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ABSTRACT

The primary bioeffect of pulsed electric fields (PEFs) is cell membrane charging, which determines a plethora of downstream events. While the kinetics of de- and hyperpolarization at the cathode- and anode-facing cell hemispheres should be the same, studies using pulsed laser strobe microscopy have reported asymmetrical charging in mammalian cells exposed to brief PEFs. To validate these anomalies, we measured charging kinetics with FluoVolt and Di-8-ANEPPS potentiometric dyes in spheroidal CHO cells exposed to 1-μs PEFs at 0–350 V/cm. Both dyes exhibited emission changes dependent on the electric field incidence angle, symmetrically at cathodic and anodic hemispheres. The changes scaled linearly with electric field strength up to at least 350 V/cm for Di-8-ANEPPS, but only to ~200 V/cm for FluoVolt. Time constants measured from the optical membrane potential kinetics did not differ between the dyes, between cathodic and anodic cell poles, or between membrane charging and relaxation. The time constants followed the theoretical dependence on cell radius for 1.6 μF/cm² specific membrane capacitance and 0.4 S/m intracellular conductance. No charging anomalies were observed. Both dyes proved suitable for characterizing nanosecond charging kinetics, with FluoVolt being more sensitive for low-noise readings, while Di-8-ANEPPS allowed for dynamic measurements within a broader range.

1. Introduction

Pulsed electric fields (PEFs) are broadly used in experimental biology and medicine for stimulating excitable cells and tissues [1–5], as well as for disrupting cell membrane barrier for intracellular drug and gene delivery or tumor ablation [6–10]. When a living cell is exposed to an external electric field, it behaves as a capacitive object: charges accumulate along the plasma membrane, and the cell interior remains essentially equipotential, so the entire voltage drop is concentrated across the thin membrane. Accordingly, PEFs polarize cells between electrodes so that the cathode-facing pole of the cell depolarizes, and the anode-facing pole hyperpolarizes.

In a spherical cell, and treating the cell membrane as a purely capacitive load, Schwan's equation describes charging as a single-exponential process, where the induced membrane potential (MP) is proportional to cell radius r and the electric field strength E :

$$MP(t) = 1.5 E r \cos\theta (1 - e^{-t/\tau}) \quad [11–14].$$

In this equation, 1.5 is a curvature factor, t is time from the onset of the pulse, θ is the polar angle from the cell center to the electric field, and τ is the charging time constant. Assuming negligible membrane conductivity, τ can be calculated as:

$$\tau = rC \left(\frac{1}{\sigma_i} + \frac{1}{2\sigma_e} \right) \quad (2)$$

where C is the membrane capacitance per unit area, and σ_i and $2\sigma_e$ are the conductivities of the intra- and extracellular compartments, respectively.

Induced MP relaxation after PEF also follows the single-exponential function, presumably with the same time constant as during charging - if the membrane has not been damaged. MP buildup and relaxation at the anodic and cathodic cell poles, disregarding the sign, can be written as:

$$MP(t) = 1.5 E r (1 - e^{-t/\tau}) \quad \text{for charging, and} \quad (3)$$

$$MP(t) = MP_{max} e^{-t_d/\tau} \quad \text{for relaxation} \quad (4)$$

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where MP_{max} is the MP reached at the end of the charging pulse, and t_d is time from the start of the discharge.

At the exact moment when PEF is turned on, the entire cell volume is exposed to the ambient electric field. As the cell membrane is charged, it shields the cell's interior from the external electric field. Once the membrane is fully charged, the induced electric field inside the cell becomes zero. In contrast, the field across the membrane is amplified profoundly by Maxwell-Wagner interfacial ionic polarization. For example, using eq. (1), a cell with a 10- μm radius in a 0.1 kV/cm electric field will charge to $\Delta V_m = 150$ mV at electrode-facing cell poles. With the membrane thickness of approximately 5 nm, the electric field across the membrane reaches $150 \text{ mV}/5 \text{ nm} = 300 \text{ kV/cm}$, which is a 3000-fold amplification of the applied PEF. This amplification is the principal reason why PEFs affect primarily cell plasma membrane, while other biological responses develop downstream of membrane polarization and permeabilization.

For typical mammalian cells, calculated τ values are on the order of 100–1000 ns [15,16]. The nearly-complete membrane polarization (to 95 % of the steady-state value) is achieved at $t = 3\tau$. This makes τ the critical parameter determining the bioeffects of brief nano- and microsecond-duration PEFs. Larger τ would prevent reaching the critical ΔV_m for electrostimulation or electroporation by brief PEFs, requiring the application of more intense electric pulses. Concurrently, cell interior will be exposed longer to the ambient electric field, potentially causing intracellular effects directly [16–18]. In elongated cells, both τ and the effective cell radius are larger when cells are aligned with the electric field than when they are placed across the field. In the aligned configuration, long, low-amplitude PEFs can charge cells to a higher ΔV_m than brief high-amplitude PEFs; however, in the across configuration, brief high-amplitude PEFs achieve a larger membrane polarization [19,20]. In excitable cells exposed to brief PEFs, τ is also the critical factor determining the balance between damage-free excitation and electroporation [21].

While knowing τ is essential for predicting and understanding many PEF effects, simple analytical solutions are valid for spherical cells only. Much more complex calculations can be extended to ellipsoidal cells [13,22], but in randomly-shaped cells ΔV_m can only be calculated through numerical simulations, which also need to be tailored to each cell's specific architecture [23]. Modeling of real cells is generally too complex and laborious to be utilized beyond proof-of-concept studies. Alternatively, τ and ΔV_m can be measured by direct experiments, which can also validate the theoretical predictions and assess possible impact of factors too complex to be included in models (such as cell membrane invaginations and the presence of intracellular organelles).

External PEFs depolarize the cathode-facing side of the cell membrane while hyperpolarizing the anode-facing side. This opposing polarization makes experimental measurements challenging, since electrophysiological methods such as intracellular microelectrodes and whole-cell patch clamp integrate the potential over the entire cell and thus will report zero net polarization. The only applicable method is the fluorescent microscopy with potentiometric dyes, which poses its own challenges: The dye's response to membrane potential should occur in less than ~ 100 ns while maintaining sufficient sensitivity to resolve small voltage fluctuations from background noise, and the MP images need to be acquired at intervals shorter than 100 ns for the detection of ΔV_m buildup during PEF.

Groundbreaking studies with quasisteady electric fields [24] confirmed the predictions of eq. (1) for $t \gg \tau$, namely the maximum polarization at electrode-facing cell poles, linear scaling of the induced MP with the electric field strength, and the azimuthal distribution of induced MP on cell membrane. The time course of membrane charging was first recorded with Di-4-ANEPPS dye in hemispheric lipid bilayers (radius 1.5 mm) [25] and with RH292 dye in urchin eggs (radius 50 μm) [26]. The latter study introduced pulsed laser fluorescence microscopy to take snapshots of the voltage response of the cell membrane at desired times into a 25- μs electric pulse. The temporal resolution of this method

does not depend on the camera speed but is determined entirely by the pulsed (stroboscopic) laser illumination. With the 300-ns laser pulse duration and the calculated 4- μs τ in a low-conductivity bath buffer, these snapshots captured gradual MP buildup at 2- μs intervals, from 2 to 10 μs into the pulse. When eggs were pulsed in a high-conductivity buffer, the maximum ΔV_m was achieved already in the first image at 2 μs into the pulse, consistent with the expected τ reduction in a more conductive medium (eq. 2). ΔV_m profiles along the cell perimeter closely followed the cosine dependence on the angle to the electric field, producing nearly symmetric charging patterns at the anode- and cathode-facing cell hemispheres.

For typical mammalian cells with τ on the order of 100–1000 ns, much shorter laser pulses and fast dyes are required to capture the charging kinetics. In 2006, Frey and co-authors [27] used strobe imaging with a 5-ns laser and Annine-6 dye to follow membrane charging in Jurkat cells exposed to 60-ns, 100 kV/cm PEFs. With this strong electric field, maximum ΔV_m was reached at 15 ns into the pulse, followed by a plateau or gradual decline, presumably because charging was halted by electropore formation. Subsequent studies with this dye, in plant cells (at 500 ns into a 1- μs pulse at 0.2–10 kV/cm [28,29]) and mammalian cells (at various times into a 60-ns pulse at 5–90 kV/cm [30]) reported profound asymmetry in charging of the cathodic and anodic cell hemispheres, even at low electric field intensities that were not expected to disrupt the membrane; the maximum ΔV_m reached at the anodic pole could be as much as 3–5 times larger than at the cathodic pole. These studies did not attempt to measure τ and the charging asymmetry came in stark contrast with earlier findings using different potentiometric dyes and longer PEFs [24–26,31,32].

In 2019, Beier et al. employed a streak camera and FluoVolt potentiometric dye to record the kinetics of membrane charging by 300- and 600-ns pulses in CHO-K1 cells [33]. The streak camera enabled a continual recording of the induced membrane potential with just 25-ns intervals, however at the expense of single-dimensional imaging and higher noise. These experiments showed a nearly symmetrical (within ~ 20 % accuracy) de- and hyperpolarization at the opposite cell poles, at least at modest non-damaging electric field strengths. Membrane charging and relaxation were slower than the electric pulse rise and fall times, consistent with the time constant “approaching 1 μs .” Interestingly, the recorded ΔV_m was about 2-fold smaller than predicted by calculations. The authors explained the difference by possible 50 % reduction of the applied voltage by “ohmic losses before the electrodes, in the electrodes, or at the electrode-solution interface.”

Building on the observation that FluoVolt dye responds to MP changes with temporal resolution of at least 25 ns [33], Kiester and co-authors employed a 6-ns pulsed laser for strobe imaging of ΔV_m kinetics in CHO-K1 cells exposed to 100-kHz, 500-kHz, and 1-MHz sine waves at 1.1–2 kV/cm [34]. Depolarization at the cathode-facing cell pole was several times smaller than hyperpolarization at the opposite pole; this difference was not explained. Measured ΔV_m followed the sine wave stimulus, but the amplitude of changes was about 3-fold less than predicted by calculations. The authors suggested that it could be caused by the membrane conductivity being much higher than estimated previously, namely at $6 \times 10^{-4} \text{ S/m}$ rather than at $3 \times 10^{-7} \text{ S/m}$.

Subsequent studies utilized strobe imaging with FluoVolt to resolve MP charging and relaxation kinetics in CHO cells (300-ns pulses at 0.16 kV/cm [35] and 600-ns pulses at 0.3 kV/cm [36]), smooth muscle cells oriented parallel and perpendicular to the electric field (300-ns pulses at 0.3 kV/cm [20]), and in variously-shaped hippocampal neurons (2- μs pulses at 0.14 kV/cm [37]). In all these experiments, fluorescence changes in response to PEFs at the anodic and cathodic poles were nearly symmetrical. In CHO and smooth muscle cells, changes often were 5–15 % larger at the anodic pole; in neurons, changes could be larger at either pole, depending on the particular cells shape and orientation to the field. Smooth muscle cells and spindle-shaped neurons charged faster when oriented perpendicular to the electric field. In neurons, charging time constants averaged $0.7 \pm 0.08 \mu\text{s}$ with no difference between the anode-

and cathode-facing membrane sites. In CHO cells exposed to 600-ns pulses at 0.3 kV/cm [36], exponential fitting of charging and relaxation kinetics yielded profoundly different time constants of 200 ns and 520 ns, respectively. This unexpected difference was duly acknowledged, yet the underlying cause remained unexamined.

Overall, studies using different potentiometric dyes, different cell types, and different recording methodologies produced remarkably different findings, some of which (such as the observed charging asymmetry) are difficult to explain. It remains unclear if the discrepancies were related to unknown dye properties, problems with dye calibration, or perhaps missing some critical factors in eqs. (1–4). The present study aimed to investigate potential charging anomalies in spheroidal mammalian cells exposed to brief 1- μ s electric pulses at sub-electroporation field strengths, and to compare experimentally measured charging time constants with theoretical predictions. We employed pulsed laser strobe imaging using two voltage sensitive dyes, FluoVolt (originally introduced as VF2.1.Cl [38,39]) and Di-8-ANEPPS [40,41], which differ significantly in their chemical structure and voltage-sensing mechanism (photoinduced electron transfer vs. electrochromism). Both dyes exhibited fast responses proportional to the electric field strength and the incidence angle. Charging and relaxation kinetics did not differ between the anodic and cathodic cell poles, and the measured time constants fell within the theoretically predicted range.

2. Materials and methods

2.1. Cell lines and media

CHO-K1 Chinese hamster ovary cells were acquired from the American Type Culture Collection (ATCC, Manassas, VA) and maintained in T25 flasks (Thermo Fisher Scientific, Waltham, MA) in Ham's F12K medium (Mediatech Cellgro, Herndon, VA, USA) at 37 °C with 5 % CO₂ in air. The medium was supplemented with 100 I.U./mL penicillin and 0.1 mg/mL streptomycin (Gibco, Gaithersburg, MD, USA), and 10 % fetal bovine serum (Atlanta Biologicals, Norcross, GA, USA). CHO cells were selected for their lack of endogenous voltage-gated channels and stable round morphology maintained for several hours after seeding [34–36].

For experiments, cells were detached from the substrate by replacing the culture medium with 0.7 ml of 0.25 % Trypsin-EDTA solution (Gibco) for approximately 3 min. Cells were resuspended in 20 ml of the culture medium, and transferred into a 35-mm glass-bottom culture dish (MatTek, Ashland, MA). The dishes were placed in the incubator for 1–3 h to ensure they attach to the glass but remain approximately spheroidal. To load cells with the potentiometric dyes (either FluoVolt or Di-8-ANEPPS, both from Thermo Fisher Scientific), the culture medium was replaced with a physiological solution composed of (in mM) 140 NaCl, 5.4 KCl, 2 CaCl₂, 1.5 MgCl₂, 10 HEPES, and 10 glucose (pH 7.2–7.3, 300–310 mOsm/kg, 1.64 S/m). The solution conductivity was measured with an ECTestr High conductivity meter (Oakton Instruments, Vernon Hills, IL) and matched the values reported previously [37,42]. FluoVolt and PowerLoad concentrate (included with FluoVolt kit) were dissolved in the physiological solution at 1:1,000 and 1:100, respectively. Alternatively, the loading solution contained 5 μ M Di-8-ANEPPS. After a 30-min incubation with the dye, cells were rinsed and the dish filled with the physiological solution was placed on the microscope stage.

2.2. Pulsed electric field treatments

Unipolar, nearly rectangular 1- μ s PEFs from an EPULSUS-FPM4–7 pulse generator [43] were delivered to selected cells via a pair of parallel tungsten rod electrodes (A-M Systems, Sequim, WA), 500 μ m in diameter and with a 1.6-mm separation. The electrodes were installed orthogonally to and at a 50- μ m height above the dish bottom using a robotic MPC-200 micromanipulator (Sutter Instrument, Novato, CA).

The microscope's field of view containing the cells was centered between the electrodes.

A TDS3052C oscilloscope (Tektronix, Beaverton, OR) captured PEF shape and amplitude through the generator's voltage and current monitor outputs. Voltage pulses were characterized over a range wider than used in the biological experiments of this study. The pulses were nearly rectangular with a rise time under 10 ns (Fig. 1A and B). Electric current increased exponentially, with the charging time constant of $\sim$ 15 ns, presumably due to the parasitic impedance of the PEF-delivering circuitry, and reached 95 % of the maximum value within $\sim$ 50 ns after PEF onset (Fig. 1C and D). To build the time course of cell membrane charging and relaxation, PEFs were delivered at 6 Hz, in synchrony with excitation laser flashes and camera shutter operation (see 2.4 for details). Current measured at the plateau of the pulse scaled linearly with the applied voltage, corresponding to the constant 120- Ω resistance of the setup (Fig. 1E).

2.3. Electric field modeling

Computational modeling of the electric field was performed with Sim4Life Lite 7.0.1.8169 software (ZMT Zurich MedTech AG, Zurich, Switzerland) as described previously [20,35,44]. The electrodes were modeled as two parallel conductive cylinders, 500 μ m in diameter and 1.6 mm apart, oriented orthogonally to the dish bottom and submerged in a 1.64 S/m solution, with their tips positioned 1.8 mm below the surface and 50 μ m above the glass coverslip (Fig. 1F). The model was meshed with approximately 19 million cells, with a minimum element size of 10 μ m. The electric field was computed in a plane 5 μ m above the glass using the Low-Frequency Electro-Ohmic Quasi-Static solver, with 1 V applied between the electrodes (Fig. 1G).

2.4. Pulsed laser strobe fluorescence microscopy

The method was described in detail in recent publications [20,34,36,37]. Cells placed on the stage of an IX71 inverted microscope (Olympus America, Center Valley, PA) were imaged with a PlanApo N 60 $\times$, 1.42 NA objective (Olympus). Both potentiometric dyes were excited by $\sim$ 6-ns, 440-nm laser flashes using a FITC fluorescence cube modified to remove residual laser harmonics at 532 nm (the excitation filter was replaced with a 533-nm notch filter NF533–17, Thorlabs Inc., Newton, NJ). Cell images were captured by an iXon Ultra 897 CCD camera with a Solis interface (Andor Technology, Belfast, UK). Laser flashes were delivered at precise time intervals with respect to the PEF using custom LabVIEW software and a DG645 delay generator (Stanford Research Systems, Sunnyvale, CA). The camera shutter opened in advance of and closed after the laser flash, capturing a single MP image per pulse.

2.4.1. Membrane polarization kinetics

Each cell was exposed to 8 trains of 70 PEFs at a constant amplitude. The excitation laser flash was delivered at different time intervals with respect to the PEF. This interval was varied in 50-ns steps, either starting 0.5 μ s prior to PEF and ending 2.5 μ s after it ("delay-increment experiments"), or starting 2.5 μ s after PEF and ending 0.5 μ s before it ("delay-decrement experiments"). Regions of interest (ROIs) were selected at cell poles facing the electrodes; they were 6 pixels wide and covered a membrane sector about 20° wide. Similarly sized ROIs were selected near the poles outside of the cell to measure the background fluorescence. The mean fluorescence intensity in the ROIs (F) was measured with MetaMorph 7.8.13 (Molecular Devices, Foster City, CA), and the background fluorescence was subtracted from all measurements. Dye bleaching by repeated laser flashes during PEF trains was compensated by averaging data from delay-increment and delay-decrement experiments, which also minimized the impact of random variations in laser flash intensity. The average of measurements before the pulse was taken as a baseline (F_0), and the optical MP changes at each timepoint were

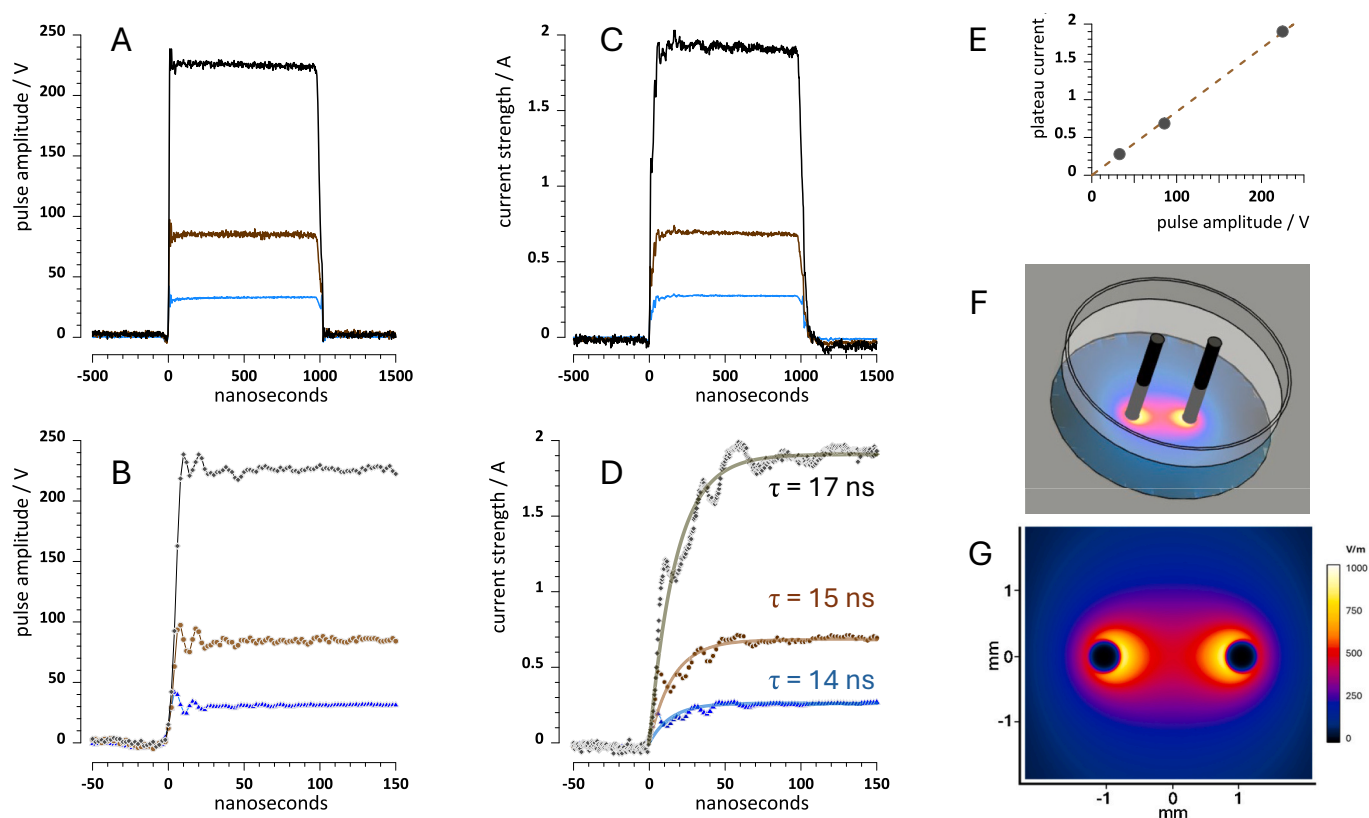


Fig. 1. Characterization of the pulsed electric field. **A:** Oscilloscope traces of 1- μ s pulses at three different voltages. **B:** The same traces at a higher temporal resolution. **C and D:** the respective traces of current between the electrodes. Traces of current in **D** were fit with a single-exponential function (solid lines), yielding charging time constant (τ) values of about 15 ns. **E:** Current at the plateau of the pulse scales linearly with the applied voltage. The linear fit through the data (dashed line) corresponds to 120 Ω resistance. **F:** The schematic of the pulsed electric field exposure setup for electric field simulations. **G:** The electric field distribution in the plane 50 μ m above the bottom of the culture dish when 1 V was applied between the electrodes. See sections 2.2 and 2.3 for more details.

expressed as $\Delta F = 100 \% \times (F - F_0)/F_0$. The resulting traces of PM charging and relaxation were fitted with exponential functions using eqs. (3) and (4), to determine the respective time constants. This was done both in individual cells and after averaging across groups of similar

sized cells.

To illustrate a representative experiment in Fig. 2, cell images were averaged across three consecutive timepoints (e.g., “100 ns” is the average of images taken at 50, 100, and 150 ns; 700 ns is the average of

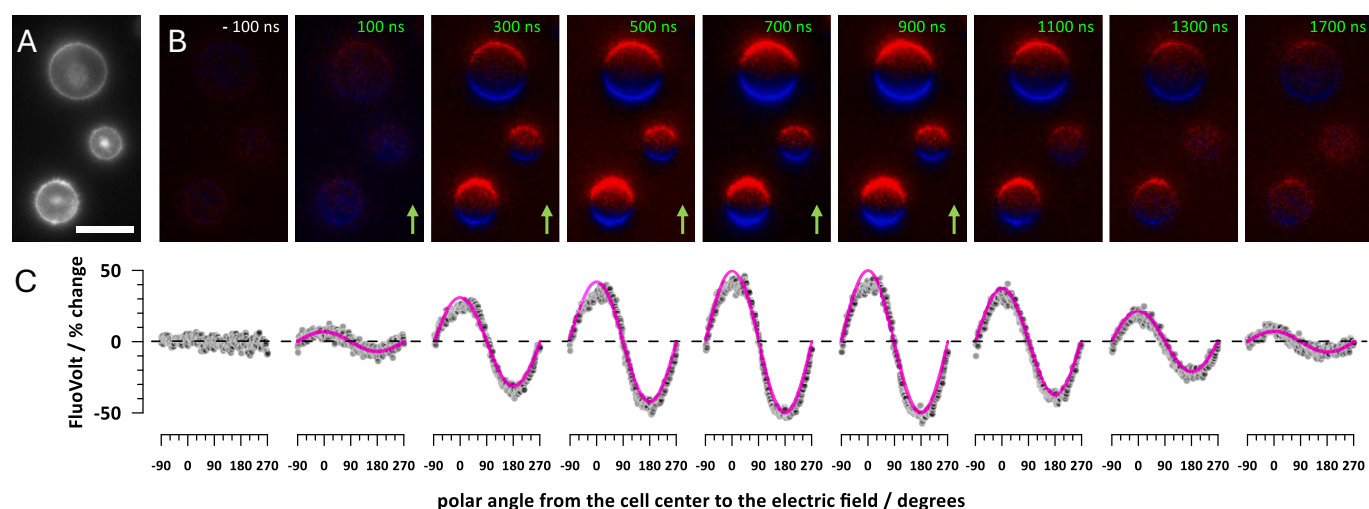


Fig. 2. Visualization of membrane charging in spheroidal CHO cells by pulsed laser strobe imaging with FluoVolt dye. **A:** Fluorescence image of three CHO cells loaded with the dye. Bar: 25 μ m. **B:** Sequential images of the optical membrane potential (MP) taken at the indicated times before, during, and after a 1- μ s, 150 V/cm pulse applied at 0 ns. Electric field direction is shown by arrows in images taken when the field was present. Increased dye emission (depolarization) is shown in red, and decreased emission (hyperpolarization) is in blue. **C:** Azimuthal charging by externally applied pulse. Optical MP was measured along the cell perimeter in the largest of the three cells, and the data were plotted against the angle from the cell center to the electric field. Solid line is the best fit of experimental datapoints to a cosine function. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

images taken at 650, 700, and 750 ns; etc.). In addition, images acquired before PEF and at late post-PEF times, when the cell had essentially discharged (after a more than a three-time-constant interval), were averaged into a single “baseline” image. Subtraction of a baseline image from a timepoint image zeroed all pixels where fluorescence decreased. The remaining pixels, where fluorescence enhanced (depolarization) were assigned red color. Similarly, subtraction of a timepoint image from the baseline one yielded only pixels with reduced fluorescence (hyperpolarization), which were assigned blue color. Color assignments and combining de- and hyperpolarization into a single image were accomplished with the “Color Align” function of MetaMorph.

To assess the dependence of cell polarization on the polar angle to the electric field, a 6-pixel-wide trace was drawn along the cell perimeter, and fluorescence intensity along this path was extracted by the “line scan” function of MetaMorph. Background subtraction and ΔF calculations were done the same way as described above. ΔF values were plotted against the angle to the electric field and fitted to a cosine function.

2.4.2. Membrane polarization dependence on PEF strength and polar angle

A total of 100 images were collected at a fixed 900-ns delay after PEF onset. The first 40 images were recorded with PEF amplitude set to zero (“sham exposures”), and the next 60 with PEF amplitude set to the desired constant value. These two sets of images were averaged to obtain the “baseline” (F) and “polarization” (F) images, respectively. Fluorescence measurements, background subtraction, and ΔF calculation were performed in a 6-pixel-wide trace drawn along the cell perimeter as described above. However, the line scan yields the number of measurements equal to the trace length in pixels, so the number of datapoints varied with the cell size and perimeter length (e.g., 260 datapoints for a smaller cell and 300 datapoints for a larger cell). To facilitate data averaging and statistical analyses despite small differences in cell size, the dataset from each cell was resampled to 360 datapoints by linear interpolation using an Excel macro, which redistributed values evenly along the normalized cell perimeter.

2.5. Data analysis and statistics

Grapher v. 16 (Golden Software, Golden, CO) and Origin 8.0 (OriginLab, Northampton, MA) were used for graph preparation and data fitting. Grapher was used to calculate linear regression, correlation coefficients, fit significance, and plot 95 % confidence intervals of the fits. It was also used to fit charging and relaxation time course data with exponential functions in eqs. (3) and (4), and to fit azimuthal dependence of polarization to a cosine function. Student's *t*-test was employed to analyze the significance of differences. Error bars are the standard error of the mean, with the number of experiments indicated in figure captions.

3. Results

3.1. Membrane charging pattern visualized by strobe imaging

In all cells tested ($n > 30$), membrane charging followed the spatial pattern and the time course predicted by eq. (1). Fig. 2A shows three representative cells stained with FluoVolt dye. Fig. 2B illustrates the changes in the dye fluorescence (i.e., the optical MP) when the cells were exposed to a 1- μ s, 150 V/cm PEF. Sequential images taken at different times during the pulse show a gradual enhancement of the FluoVolt signal (depolarization) in the cathode-facing cell hemisphere and the opposite effect (decreased fluorescence and hyperpolarization) in the anode-facing hemisphere. The azimuthal distribution of the induced MP was analyzed in the largest of the three cells and fitted to a cosine function of the incidence angle (Fig. 2C). All the fits were highly significant ($p < 0.0001$), with correlation coefficients ranging from 0.87 for noisier weak effects (at 100 ns and 1700 ns) to 0.98 when MP changes

clearly exceeded the noise (at 700 and 900 ns). Polarization of the cathode- and anode-facing hemispheres was nearly symmetrical and peaked at the poles. It reached apparent saturation at 700 ns into the pulse, and dropped by 85 % at 700 ns after it.

In two smaller cells in Fig. 2B, depolarization was visually more pronounced than hyperpolarization. While such asymmetry was sometimes observed in individual cells, below in 3.3 we show that data averaged across the cell population shows no systematic difference between the opposite cell poles.

3.2. Effect of the electric field strength

Fig. 3A displays the azimuthal distribution of changes in the emission of FluoVolt (top) and Di-8-ANEPPS (bottom) at the end of 1- μ s PEFs applied at the electric field strength from 43 to 357 V/cm. Experimental data showed strong correlation (>0.9 for most panels) with best fits using a cosine function. Peak polarization measured at the cathodic and anodic poles (Fig. 3B) increased linearly with the electric field strength, with the exception of FluoVolt data above 200 V/cm. Within the linear range, FluoVolt exhibited more than a 2-fold higher sensitivity to the external electric field than Di-8-ANEPPS (27 ± 1 % vs 10.8 ± 0.4 % change per 100 V/cm).

The reduced FluoVolt response at higher electric field strengths could potentially be caused by membrane permeabilization, which would reduce charging efficiency, thus yielding a smaller MP buildup than expected. The cells used in Fig. 3 experiments with FluoVolt had the average radius of 8.7 ± 0.7 μ m; using eq. (3) and assuming $t > \tau$, the MP induced by the 200 V/cm electric field at cell poles could reach ~ 250 mV, which, at least with longer electric pulses, might cause electroporation [32,45–48]. At the same time, the response of Di-8-ANEPPS remained linear, indicating that electroporation either did not happen or did not hinder charging, and the non-linearity of FluoVolt was caused by approaching its limits of voltage sensitivity. Further experiments were limited to electric fields at intensities within the linear range of the dyes.

3.3. Time kinetics of membrane charging and relaxation: Effect of cell size, electric field strength, and dye selection

Fluorescence images of cell membrane were taken at 50-ns intervals before, during, and after a 1- μ s pulse, and induced optical MP values (ΔF , %) at the cathode- and anode-facing cell poles were plotted against time (Fig. 4). Measurements averaged across several cells of a similar size (Fig. 4A) or having a similar size but exposed at different field strengths (Fig. 4B) were fitted with exponential functions using eqs. (3) and (4). The *P*-value of all fits in Fig. 4 was less than 0.001, with the correlation coefficient higher than 0.9.

In FluoVolt-loaded cells, charging at 90 V/cm was determined by the cell radius: smaller cells charged faster but to a smaller limit MP than larger cells (Fig. 4A). For instance, the charging time constant and the limit MP (both measured from the best fit equations) at the anodic pole of cells with 6.9 ± 0.15 μ m radius were 283 ns and 13 %, respectively. For larger, 8.2 ± 0.15 μ m cells, these values increased to 353 ns and 20 %; and in still larger 11.7 ± 0.7 μ m cells they reached 418 ns and 28 %. MP kinetics at the cathodic pole and the relaxation time constants followed the same dependence on cell size.

Previous studies with FluoVolt established that this dye is sufficiently fast to detect nanosecond-timescale MP changes [33,34]. While Di-8-ANEPPS is commonly referred to as a “fast potentiometric dye”, this definition refers to its ability to follow millisecond-scale MP changes, such as action potentials and membrane charging by millisecond electric pulses [40,41,49,50]. The suitability of Di-8-ANEPPS for recording nanosecond-scale processes has not been reported. We compared charging kinetics as monitored with FluoVolt and Di-8-ANEPPS dyes (Fig. 4A and B) and did not observe any difference. Charging and relaxation time constants were nearly the same when measured with

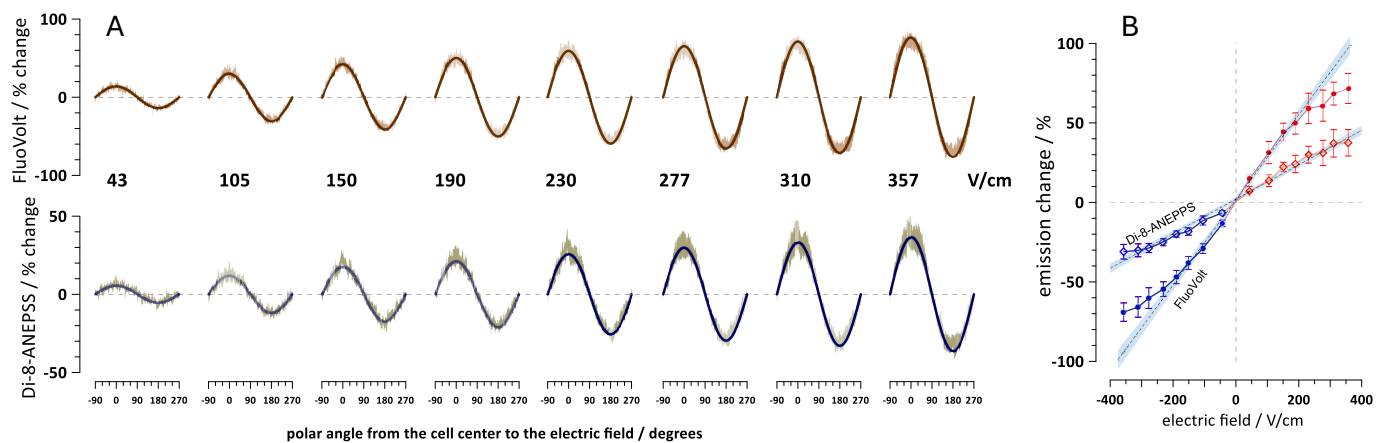


Fig. 3. Effect of the electric field strength on membrane polarization at 900 ns into a 1- μ s pulse. **A:** Azimuthal distribution of the optical MP in cells loaded with FluoVolt dye (top panels, $n = 4$, cell radius $8.7 \pm 0.7 \mu\text{m}$) and with Di-8-ANEPPS dye (bottom panels, $n = 4$, cell radius $9.5 \pm 0.9 \mu\text{m}$). Solid lines are the best fit to a cosine function; shadows in a lighter color are the standard error bars for 360 datapoints in each graph; the mean values are not shown for clarity. Field strength (V/cm) is indicated between the panels. Note different vertical scales for the two dyes. **B:** Effect of the electric field strength on de- and hyperpolarization at anode- and cathode-facing cell poles (blue and red symbols, respectively). Polarization was measured within $\pm 5^\circ$ sectors from the poles in FluoVolt- and Di-8-ANEPPS-loaded cells (filled and open symbols, respectively; mean $\pm$ s.e., $n = 4$). For convenience of fitting, the electric field was considered negative for the anodic pole data and positive for the cathodic pole data. Dashed lines are the best linear fits, and shaded areas are the 95 % confidence intervals of the fits. For FluoVolt, fitting was limited to the data range from -200 to 200 V/cm but plotted up to the axis limits. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

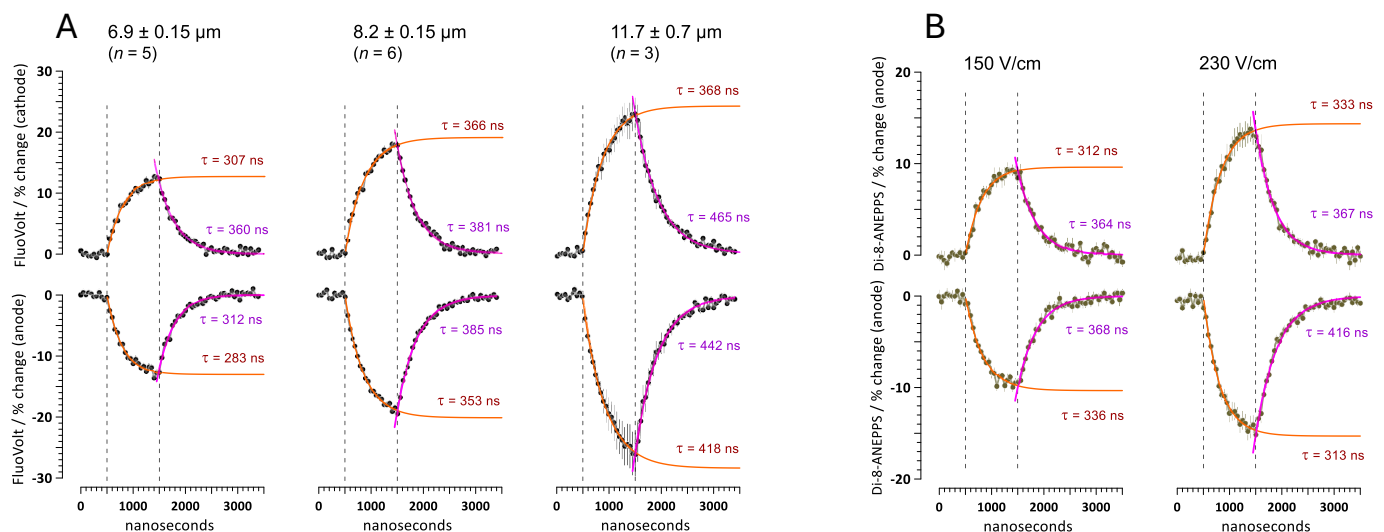


Fig. 4. The kinetics of membrane charging and relaxation in spheroidal CHO cells subjected to a 1- μ s electric pulse. **A:** Cells were loaded with FluoVolt and exposed at the same field strength of 90 V/cm. Cells with a similar radius (legends, μm) were grouped together, and the kinetics of polarization were plotted separately for the cathodic and anodic cell poles (mean $\pm$ s.e.). Pulse delivery time is marked by vertical dashed lines (from 500 to 1500 ns). Solid lines are the best fits of charging and relaxation using eqs. (3) and (4), respectively (see text). The time constant values (τ) measured from the fits are labeled next to the fits. **B:** A total of 5 CHO cells with a similar radius ($7.8 \pm 0.2 \mu\text{m}$) were loaded with Di-8-ANEPPS and exposed at the field strengths of 150 and 230 V/cm. Other details are the same as for panel A.

either dye in cells of the same radius, proving that Di-8-ANEPPS is sufficiently fast to measure cell charging kinetics. Measured constants were not affected by the electric field strength when it was within the linear range of the dye response. Small differences in the constants between 90, 150, and 230 V/cm in Fig. 4 are an expected result of data dispersion affecting the regression, and are not indicative of any systematic or reproducible effect.

3.4. The symmetry of membrane charging and measured versus theoretical time constants

Taking into account previous reports of asymmetric charging kinetics when mammalian and plant cells were exposed to brief PEFs [27–30,33,34], even at a non-electroporating electric fields strength, we

compared the time constants at the anode- and cathode-facing cell poles in individual cells (Fig. 5A), as well as charging and relaxation time constants at the same cell pole (Fig. 5B). We started with data presented in Fig. 4A but measured charging kinetics for each individual cell separately (including cells of intermediate sizes that were not included in Fig. 4A). The data were fitted with exponential functions (similarly to Fig. 4), to yield single-cell time constants τ for charging and relaxation at cathodic and anodic poles. Ratios of these values were plotted against the cell radius in Fig. 5A and B. All but two individual ratio values fell in the range between 0.6 and 1.5, and deviations from unity were the result of fitting noisy data rather than a reflection of actual differences. Overall, there was no difference between charging τ at cathodic and anodic poles. The ratios of cathodic-to-anodic time constants, averaged across all cells regardless of their size, were 0.99 ± 0.06 for charging and

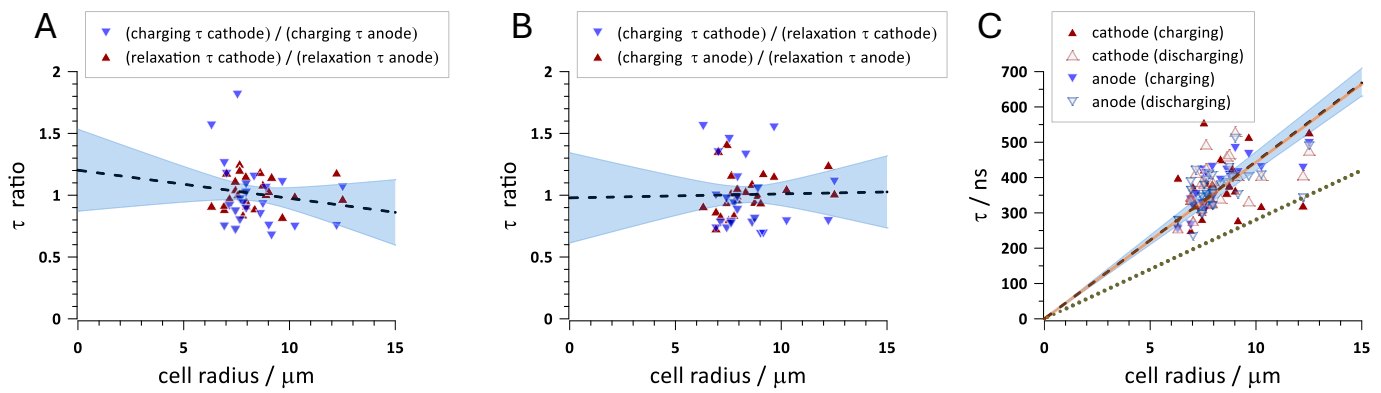


Fig. 5. The lack of difference between the time constants (τ) for membrane charging and relaxation (A) or at the different cell poles (B) and their comparison with theoretical values (C). A: Ratios of time constants at the opposite poles in 24 cells are plotted against the cell radius, separately for charging and relaxation. The dashed line and the shaded area are the best linear fit through the data and its 95 % confidence interval. B: Ratios of charging and relaxation time constants in 24 cells are plotted against the cell radius, separately for the cathodic and anodic poles. The dashed line and the shaded area are the best linear fit through the data and its 95 % confidence interval. C: All time constants ($n = 96$, 4 values per cell) were fit to a linear regression through the origin (the dashed line); the shaded area is a 95 % confidence interval. The dotted line is a theoretical dependence of the time constant on cell radius, calculated using eq. (2) for the internal cell conductivity of 0.4 S/m and the specific membrane capacitance of 1 $\mu\text{F}/\text{cm}^2$. Increasing the latter value to 1.6 $\mu\text{F}/\text{cm}^2$ yields a steeper dependence (semi-transparent solid brown line) that is identical to the best fit line. See text for more details. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

1.03 ± 0.03 for relaxation ($n = 24$). Likewise, there was no difference between τ values for charging and relaxation (0.985 ± 0.06 and 1.025 ± 0.04 for cathodic and anodic poles, respectively; $n = 24$). The slope of linear fits through the data was not significantly different from 1 (-0.023 ± 0.02 in Fig. 5A and 0.003 ± 0.02 in Fig. 5B), indicating that the ratios of time constants did not depend on cell size.

All measured τ values (96 datapoints from 24 cells) were plotted against the cell radius and fitted to a linear regression through the origin (Fig. 5C), yielding the slope coefficient of 44.5 ns/ μm . A theoretical curve using extracellular conductivity of 1.64 S/m [42], intracellular conductivity of ~ 0.4 S/m in CHO cells (range: 0.37–0.42 S/m in different studies [51]), and the specific membrane capacitance of 1 $\mu\text{F}/\text{cm}^2$ (a “typical” value assumed across multiple studies, e.g. [42,52–54]) yields a comparable but smaller slope of 28 ns/ μm (dotted line in Fig. 5C). However, increasing the specific capacitance in eq. (2) from 1 to 1.6 $\mu\text{F}/\text{cm}^2$ yields a line that matches the linear regression curve to experimental data. While we could not find in published literature the specific membrane capacitance for CHO cells, the values measured in other cell types by methods such as electrorotation and impedance spectroscopy often were in the same range (e.g., 1.53 ± 0.43 $\mu\text{F}/\text{cm}^2$ in monocytes [55], 1.94 ± 0.14 $\mu\text{F}/\text{cm}^2$ in normal mammary epithelial MCF-10 A cells [56], 1.86 ± 0.11 $\mu\text{F}/\text{cm}^2$ in human breast adenocarcinoma cells [56], and 1.82 ± 0.24 $\mu\text{F}/\text{cm}^2$ in DS19 murine erythroleukemia cells [57]; see [58] for more examples).

4. Discussion

While pulsed laser photography of the induced optical MP was introduced over two decades ago, only two previous studies [36,37] employed strobe imaging to determine membrane time constants directly from charging and relaxation kinetics. The challenges encountered by such measurements include multiple repeated exposures to PEFs and accumulation of kinetic data at nanosecond resolution while avoiding cell damage; accounting for laser variability and dye bleaching; and staying within the linear range of the potentiometric dye response. Our previous study measured time constants in neurons [37]; due to their irregular shape, strobe imaging was the only feasible method. These data helped establish mechanisms and frequency limits of neurostimulation by sinusoidal stimuli. However, a question that remained unanswered was whether time constants measured by strobe imaging are comparable to theoretical values calculated from cells’ and media dielectric properties. A number of previous studies with FluoVolt and

Annine-6 dyes suggested that significant discrepancies could take place. Among those, studies with Annine-6 consistently reported distinctly different charging kinetics at the cathode- and anode-facing poles in spherical cells [27–30]. Such asymmetry could not be explained by charging theory and eq. (1), thus implying either an unknown property of mammalian and plant cells exposed to sub-microsecond PEFs, or an unforeseen behavior of Annine-6 dye. MP imaging studies with FluoVolt dye reported a different anomaly, namely a much smaller induced MP than predicted by eq. (1) from the electric field strength [33,34]. This result suggested that the cell membrane is 2000-fold more conductive than determined by dielectric measurements [34]. In a separate study [36], membrane relaxation was reported to be 2.5 times slower than charging, a result not accounted for by purely capacitive behavior.

In contrast, we have not observed any charging anomalies with either FluoVolt or Di-8-ANEPPS dye. We deliberately avoided voltage calibration of either dye, which could introduce an additional error, especially in case of extrapolation to suprphysiological MPs where the dye sensitivity becomes uncertain. The non-linearity of FluoVolt response beyond ± 200 mV could be caused by either membrane electroporation or by approaching the saturation of the dye response, or both. Concurrent measurements with Di-8-ANEPPS dye remained linear beyond ± 200 mV, proving that the cell membrane can withstand such voltages, and the non-linearity of FluoVolt was likely due to the response saturation. However, one cannot exclude that FluoVolt sensitized the membrane to electroporation by weaker PEFs, making FluoVolt-stained cells susceptible to electroporation at voltages that would not disrupt membrane in unstained and Di-8-ANEPPS-stained cells.

Our experiments were performed with 1- μs PEFs, whereas most studies reporting asymmetric membrane charging used shorter pulses, from 60 to 600 ns [27–30,33,36]. Nonetheless, we infer that our conclusions extend to shorter PEFs as well. The optical MP response to the applied electric field was first recorded at 50 ns time into the pulse, and at 50-ns intervals thereafter (Fig. 4). The MP reached at any of these early timepoints did not differ between the anodic and cathodic poles, providing evidence for symmetrical charging by any PEFs from 50 ns to 1 μs in duration. Likewise, the time constant would not change appreciably if calculated from partial fits to the experimental data (e.g., from 0 to 300 ns or to 600 ns, instead of 1 μs). Anomalous charging kinetics reported previously [27–30,33,36] could be a combined effect of electroporative membrane disruption and the uncertainty of dye calibration. Notably, the asymmetry of the optical MP expressed as the dye emission ratio was much smaller than when it was converted into volts using a

theoretical calibration curve [27,30]. This calibration was based on the Stark shifts measured between -70 mV and $+10$ mV in neurons [59], and its accuracy and applicability, particularly at voltages far outside this range, could be compromised by factors unaccounted for (different cell type, dye bleaching, internalization, excitation by a pulsed laser, solvatochromism, etc.)

Our results highlight the need to consider multiple caveats associated with the quantitative use of potentiometric dyes and with pulsed laser detection of charging kinetics. With all factors taken into account, strobe imaging in mammalian cells produces reliable results that match theoretical predictions and dielectric measurements by non-optical methods. This conclusion enables the extension of strobe imaging to random-shaped cells and cell groups, where other methods and modeling may not be applicable.

CRediT authorship contribution statement

Iurii Semenov: Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Giedre Silkuniene:** Writing – review & editing, Formal analysis, Data curation. **Mantas Silkunas:** Writing – review & editing, Investigation. **Olga N. Pakhomova:** Writing – review & editing, Investigation. **Joel N. Bixler:** Resources, Methodology, Investigation. **Bennett L. Ibey:** Writing – review & editing, Methodology, Investigation. **Mark A. Keppler:** Software. **Andrej G. Pakhomov:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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